

Heard v0.1 → v0.2: A Korean Long-Term Memory Benchmark and On-Device Retrieval Pipeline for Solo-Business Monologue

Chanyoung Kim · NLP Term Project Part 2 · 2026-04-28

Website	https://cykim05.github.io/
Code	https://github.com/cykim05/heard
Dataset	https://huggingface.co/datasets/chanyoungkim/heard-bench

Summary

Heard (Part 1 NLP proposal) is an on-device Korean LLM assistant that reflects a user's past self back at them at decision moments. Its three architectural pillars are MIC (speech-to-text, implemented as a sounddevice + faster-whisper pipeline but not evaluated in v0.1), NODE (domain-specific typed memory), and MIRROR (reflective response). Part 2 asks whether a lightweight on-device memory pipeline yields measurable gains for a small Korean LM on nightly-self-talk questions, and whether a reflective prompt policy beats a generic advisory one.

Evaluating this requires a long-term memory benchmark in Korean, covering monologue (not dialogue), in the solo-business domain. No existing benchmark — LongMemEval, PerLTQA, or LoCoMo — satisfies all three criteria, so we constructed heard-bench: 270 items across three tracks (en_subset, ko_translated, ko_native), generated from a 2,046-utterance synthetic corpus over three solo-business personas, adversarially filtered against no-memory baselines, 4-gate auto-validated, and author-reviewed. The dataset is released on HuggingFace under CC-BY-4.0.

On the resulting benchmark, the v0.2 sweep evaluates eleven SUT configurations across four model families (Kanana 1.5, Qwen 2.5, HyperCLOVA-X SEED, and the Korean-tuned Llama 3 derivatives Blossom and Open-Ko), three parameter scales (1.5–8 B), and a 4-bit-NF4 quantization axis applied wherever the model fits within a 30 GB memory budget. The sweep totals 14,850 SUT generations and 2,112 pairwise judge verdicts.

Three results are robust across the lineup. First, dense retrieval lifts contains-pass on **ko_native** for every configuration (mean +11.8 pp **retrieval-no_node**, minimum +6.2 pp, no negative deltas), with the on-device 2.1 B target named in Part 1 (**kanana_nano**) following the canonical 4.7 % → 10.9 % → 15.6 % trajectory. Second, the reflective policy dominates the advisory baseline on every rubric across 528 pairwise judge decisions per rubric, with the two "mirror, do not advise" rubrics — emotional attunement and open-question framing — landing at 88.1 % and 88.3 % reflective share. Third, the language-axis ordering $\text{en_subset} \leq \text{ko_translated} < \text{ko_native}$ is preserved by ten of eleven SUTs, establishing Korean-native data as necessary rather than optional. The strongest single configurations on **ko_native** retrieval are **kanana_nano_int4** (20.3 %, but borderline-noise under paired analysis), **kanana_8b** and **kanana_8b_int4** (18.8 %), and **open_ko_8b** (17.2 %); combining retrieval pass, reflective share, latency,

and quantization sensitivity, we recommend **kanana_8b** as the v0.2 default deployment SUT for the *Heard* product.

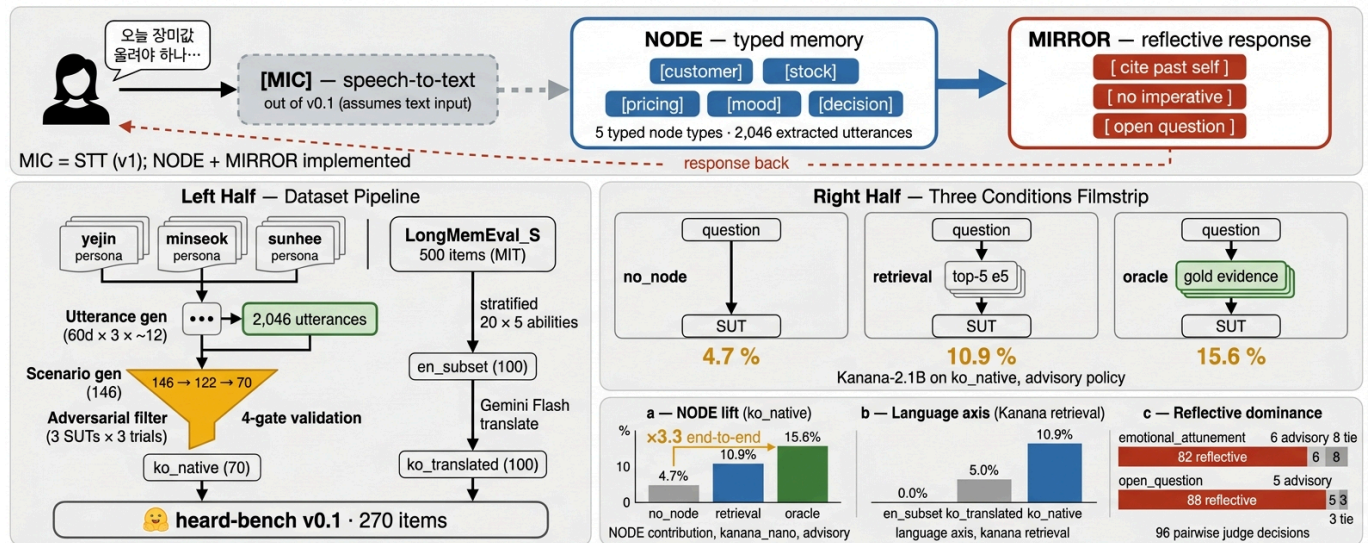


Figure 1: Heard v0.1 Pipeline & Headline Result Overview

{#fig:overview}

Figure 1. *Heard v0.1 pipeline and headline results at a glance.* The **top band** shows the three-pillar architecture — MIC for speech-to-text (dashed, not evaluated in v0.1), NODE for the five typed node categories (customer / stock / pricing / mood / decision), and MIRROR for the reflective response style (cite past self, no imperative, open question). The **middle band** is split into two halves. The left half compresses the dataset construction pipeline: three persona cards and the LongMemEval_S source feed into a 2,046-utterance corpus and a 146 → 122 → 70 ko_native funnel, converging on heard-bench v0.1 (270 items). The right half is a three-condition filmstrip that shows what each condition delivers to the SUT: question only (**no_node**, 4.7%), question + top-5 retrieved (**retrieval**, 10.9%), and question + gold evidence (**oracle**, 15.6%). The **bottom band** summarises the three headline findings — the NODE lift on **ko_native** (×3.3 end-to-end for Kanana), language-axis decay across tracks, and the reflective-vs-advisory judge outcomes on emotional attunement (82 wins) and open-question framing (88 wins) out of 96 pairwise decisions.

1 Introduction

1.1 Background

Heard is the Part 1 NLP proposal for an on-device Korean LLM assistant. Its architecture has three pillars. MIC handles always-off tap-to-talk speech-to-text; it is implemented in v0.1 but not evaluated in Part 2 (§2.1). NODE is the domain-specific typed memory built over utterances. MIRROR produces reflective responses that quote the user's past words and end with an open question rather than giving advice.

The proposal targets Korea's 1.4M-member solo-business community, where nightly self-talk arrives from people who have no coworkers, no staff, and no one to check in with. Part 2 asks two empirical questions: (1) does a lightweight retrieval-based memory pipeline yield measurable gains for a small Korean on-device LM on nightly self-talk questions, and (2) does a reflective prompt policy beat a generic advisory policy when memory is available?

1.2 The benchmark gap

Answering these empirical questions requires a long-term memory benchmark meeting three criteria: (a) Korean, (b) monologue (nightly self-talk, not dialogue), and (c) solo-business decision-making. A survey of recent long-term memory benchmarks returns no candidate that satisfies all three.

Benchmark	Language	Format	Domain
LongMemEval (Wu et al., ICLR 2025)	EN	user–AI dialogue	generic chat
PerLTQA (Du et al., 2024)	ZH / EN	human–human dialogue	social events
LoCoMo (Maharana et al., 2024)	EN	long dialogue	generic personal

We therefore construct **heard-bench**: a 270-item benchmark across three tracks. The construction pipeline is: (1) generate a 2,046-utterance synthetic corpus over three solo-business personas (florist, roaster-café owner, one-chair hair salon), (2) compose scenarios with today-questions and evidence utterances, (3) adversarially filter against no-memory baselines (3 SUTs × 3 trials, keeping only items where all fail), (4) 4-gate auto-validate (evidence consistency, history-only fail, cross-generator duplication, ambiguity score), and (5) author-review the survivors for naturalness and gold-answer correctness. The dataset is released on HuggingFace under CC-BY-4.0.

1.3 Contributions

- 1. Dataset construction as necessary infrastructure** (§2.2). We construct, adversarially filter, 4-gate validate, and author-review heard-bench, a 270-item set across en_subset, ko_translated, and ko_native tracks, and release it on HuggingFace under CC-BY-4.0. No existing long-term memory benchmark meets our three criteria of Korean / monologue / solo-business.
- 2. Empirical NODE lift** (§3.1). Dense retrieval lifts Kanana-2.1B's ko_native pass rate from 4.7% (no memory) to 10.9% (retrieval) to 15.6% (oracle), a 3.3× end-to-end gain over the memoryless baseline.
- 3. Language axis decay confirms Korean-native necessity** (§3.2). Kanana's retrieval pass rate decays monotonically: ko_native 10.9% → ko_translated 5.0% → en_subset 0.0%. Korean-native data is necessary, not optional, for a Korean on-device assistant.
- 4. Reflective policy dominates on emotional and open-question axes** (§3.3). Across 96 pairwise judge decisions per rubric, reflective beats advisory on emotional attunement 82/96 and on open-question framing 88/96, supporting the Part 1 MIRROR thesis.

2 Methods

All code, configurations, and experiment artifacts are available at <https://github.com/cykim05/heard> (Apache 2.0). The dataset is published at <https://huggingface.co/datasets/chanyoungkim/heard-bench> (CC-BY-4.0). Per-call API logs under `experiments/_api_log/` and per-run folders under `experiments/<run_id>/` enable full reproducibility; diskcache keyed on (model, messages, seed) makes a cold rerun free.

2.1 MIC — recording and transcription

Heard's first pillar turns spoken nightly monologue into text before NODE ingests it. We implement it as a lightweight tap-to-talk recorder plus a Whisper-family transcriber; the code ships with v0.1 but is not exercised by the evaluation in §3.

Recorder (`src/mic/recorder.py`). A `sounddevice` wrapper captures 16 kHz mono PCM in float32, which is the format Whisper-family models expect natively. Two usage modes are supported: `record_fixed(seconds)` for scripted capture, and a `Recorder` context manager for tap-to-talk (start, wait for the user, stop). The captured array is either passed directly to the transcriber or written to a 16-bit WAV via `save_wav`.

Transcriber (`src/mic/transcribe.py`). A thin `faster-whisper` wrapper (CTranslate2 backend), defaulting to the Whisper `small` model (244 M parameters). Language is fixed to Korean; VAD filtering and beam search are exposed as kwargs. Heavy model loading is lazy, so importing the module is cheap even in environments that never run STT. `transcribe()` returns a single concatenated transcript; `transcribe_segments()` returns per-segment timestamps for later utterance splitting.

Integration. `scripts/00_record_and_transcribe.py` is the reference integration: it records via `Recorder`, optionally saves a WAV, and routes the audio through `Transcriber`. The script ships as a working demo rather than a v0.1 evaluation target, since measuring WER on Korean casual monologue requires a dedicated audio benchmark that we leave to v2. All subsequent evaluation in §3 operates on gold utterance text so that the memory-pipeline question is isolated from STT error.

2.2 Dataset construction

The heard-bench construction pipeline has five stages: corpus generation, scenario composition, adversarial filtering, 4-gate auto-validation, and author review. We describe each in turn.

Personas. Three synthetic Korean solo-business owners anchor the corpus: Yejin (florist, Mapo), Minseok (roaster-café, Seongsu), and Sunhee (one-chair hair salon, Hongje). Each persona card specifies regulars, stock or services, recurring stressors, and timestamped historical anchors (`days_ago`-indexed past events) so that temporal-reasoning and multi-session items have grounded evidence. The persona set is synthetic to avoid privacy concerns; all names and establishments are fictitious.

Utterance corpus. One generator call per (persona, day) block across 60 days produces 12 utterances per day on average, rotating across `anthropic/claude-haiku-4.5`, `openai/gpt-4o-mini`, and `google/gemini-2.5-flash` so that no single generator's fingerprint dominates. 43% of the resulting utterances reference a persona historical anchor, and these references become retrieval gold hooks in later stages.

Persona	Role / location	Days	Utterances	Historical-ref share
yejin_florist	Florist, Mapo	60	686	43%
minseok_cafe	Roaster-café, Seongsu	60	687	43%
sunhee_hair	One-chair hair salon, Hongje	60	673	43%
Total			2,046	881 / 2,046

Three tracks. heard-bench is organized into three tracks.

Track	Items	Language	Haystack	Source	License
en_subset	100	EN	EN	LongMemEval_S stratified (20 per ability)	MIT

Track	Items	Language	Haystack	Source	License
ko_translated	100	KO Q + answer	EN (preserved)	Gemini 2.5 Flash translation	CC-BY-4.0
ko_native	70	KO	KO (our corpus)	persona-driven gen + adversarial filter	CC-BY-4.0
Total	270				

Ability distribution (ko_native). The native-track items are stratified by the six long-term-memory abilities we target. IE, MR, KU, TR, and ABS follow LongMemEval's taxonomy; REFL (reflective quality) is specific to Heard.

Ability	Count	What it tests
IE	19	Specific fact recall
TR	15	Temporal reference resolution
KU	13	Knowledge update (latest state)
ABS	10	Abstention (never mentioned)
MR	7	Multi-session aggregation
REFL	6	Reflective response quality (LLM-judged)

Validation pipeline. ko_native candidates pass through four stages before inclusion:

- 1. **Adversarial filtering:** 150 candidates generated → 146 successfully parsed → 122 survived (all three no-memory baselines, Kanana-2.1B / Qwen-1.5B / Qwen-3B, fail across three trials each). This ensures that memory is structurally necessary to solve the retained items.
- 2. **4-gate auto-validation:** evidence-answer consistency, history-only fail reproduction, cross-generator 4-gram overlap < 15%, and ambiguity score ≥ 3/5 from a third-party LLM rater (ADR 0004).
- 3. **Author review:** the 70 surviving items were reviewed by the author for naturalness, evidence alignment, and gold-answer correctness. Minor edits to gold substrings and fluency fixes were applied as needed.
- 4. **Final count:** 70 ko_native items, shorter than the originally planned 100. The rejection rate at Gate 4 is discussed in §4.5.

Algorithm 1 summarises the adversarial filter. Any candidate that is solvable without memory by any of the three SUTs in any of three trials is discarded; only candidates where memory is structurally required survive.

```
Algorithm 1 – Adversarial filter
-----
input  : candidate set C, SUT pool M = {m1, m2, m3},
        trials T = 3
output : hard subset H ⊆ C
```

```
H ← ∅
for each candidate c ∈ C:
  any_pass ← False
  for each m ∈ M:
    for t in 1..T:
      r ← m.generate(c.question)      // no memory
      if gold_criterion(r, c.gold):
        any_pass ← True
        break
    if any_pass: break
  if not any_pass:
    H ← H ∪ {c}
return H
```

ADR 0003 summary. Track B keeps only question and gold_answer in Korean; haystack stays English, because full-session translation exceeded Gemini-Flash's output token cap. Chunked-session translation is a concrete v2 path.

ADR 0004 summary. Gate 1 was demoted to advisory after a scenario-generation regression produced false negatives that did not correlate with SUT evaluation value.

2.3 Memory pipeline

Retriever. We use [intfloat/multilingual-e5-small](#) (118 M parameters), mean-pool the final hidden states, L2-normalize, and take cosine top-5. The index is per-persona for ko_native (~680 utterances each) and per-item for en_subset and ko_translated (~50 sessions per item).

2.4 Systems under test

The v0.1 sweep evaluated two SUTs ([kanana_nano](#) and [qwen25_3b](#), both fp16), and the v0.2 expansion this report describes runs the strict superset shown in Table A. The two v0.1 rows are retained verbatim in the table for continuity; reading the table top-to- bottom recovers the v0.1 sweep, while reading all eleven rows gives the v0.2 sweep. All configurations run sequentially on a single NVIDIA L40S 48 GB, with the embedder and the SUT co-resident in memory; the listed memory budget for int4 is the observed peak during the sweep run.

Table A. SUT lineup (v0.2 = strict superset of v0.1). Configurations marked v0.1 are the two-SUT reference set from the v0.1 release (results archived under [experiments/20260423_1610_day3_sweep/](#)); all eleven rows are evaluated in §3.1–§3.5 of this report.

Logical	HF id	Params	Quant	License	Sweep
kanana_nano	kakaocorp/kanana-1.5-2.1b-instruct-2505	2.09 B	fp16	Apache-2.0	v0.1, v0.2
qwen25_3b	Qwen/Qwen2.5-3B-Instruct	3.09 B	fp16	Apache-2.0	v0.1, v0.2
kanana_nano_int4	kakaocorp/kanana-1.5-2.1b-instruct-2505	2.09 B	int4	Apache-2.0	v0.2

Logical	HF id	Params	Quant	License	Sweep
kanana_8b	kakaocorp/kanana-1.5-8b-instruct-2505	8.03 B	fp16	Apache-2.0	v0.2
kanana_8b_int4	kakaocorp/kanana-1.5-8b-instruct-2505	8.03 B	int4	Apache-2.0	v0.2
hclova_seed_15b	naver-hyperclova/HyperCLOVAX-SEED-Text-Instruct-1.5B	1.59 B	fp16	HCX-SEED-Public (gated)	v0.2
hclova_seed_15b_int4	naver-hyperclova/HyperCLOVAX-SEED-Text-Instruct-1.5B	1.59 B	int4	HCX-SEED-Public (gated)	v0.2
qwen25_3b_int4	Qwen/Qwen2.5-3B-Instruct	3.09 B	int4	Apache-2.0	v0.2
qwen25_7b	Qwen/Qwen2.5-7B-Instruct	7.62 B	fp16	Apache-2.0	v0.2
bllossom_8b	MLP-KTLim/llama-3-Korean-Bllossom-8B	8.03 B	fp16	Llama-3 Community	v0.2
open_ko_8b	beomi/Llama-3-Open-Ko-8B-Instruct-preview	8.03 B	fp16	Llama-3 Community	v0.2

The v0.2 sweep extends the v0.1 lineup along three axes: a third parameter decade ([hclova_seed_15b](#) at 1.5 B and [kanana_8b](#) at 8 B), a model-architecture axis (the Korean-tuned Llama 3 derivatives [bllossom_8b](#) and [open_ko_8b](#)), and a 4-bit-NF4 quantization axis on the four configurations that fit under a 30 GB memory budget. HyperCLOVA-X SEED 1.5B was admitted once HuggingFace gated-access approval was granted. Three further SUTs were attempted but dropped: [LGAI-EXAONE/EXAONE-3.5-2.4B-Instruct](#) (the installed `transformers` 4.57.6 lacks the `RopeParameters` symbol the EXAONE config imports), [yanolja/EEVE-Korean-Instruct-10.8B-v1.0](#), and [upstage/SOLAR-10.7B-Instruct-v1.0](#) (fp16 throughput too low under the reflective policy to finish within the wall-clock budget, ≈ 47 and ≈ 200 generations per hour respectively); all three remain v0.3 candidates.

Per-family description. Four model families participate in the v0.2 sweep, each chosen for a specific role.

- **Kakao Kanana 1.5 (Apache-2.0).** Kanana-1.5-2.1B is the primary on-device target for the *Heard* product and the SUT whose v0.1 numbers anchor §3.1; the 8B sibling Kanana-1.5-8B is included as an in-family reference ceiling. Both are Korean-native instruction-tuned models from Kakao, available under a permissive Apache-2.0 license that does not restrict redistribution of model output.
- **NAVER HyperCLOVA-X SEED (HCX-SEED-Public, gated).** HyperCLOVAX-SEED-Text-Instruct-1.5B is NAVER's smallest Korean-native instruction model and the lightest configuration in the lineup at 1.59 B. Its license requires HuggingFace gated-access approval, which was granted to the runtime account on 2026-04-26; outputs are retained in [experiments/](#) and the model is not redistributed by this report.
- **Alibaba Qwen 2.5 (Apache-2.0).** Qwen 2.5-3B and Qwen 2.5-7B are multilingual instruction-tuned baselines, included to measure how much a strong general multilingual model can close the gap against a Korean-native model on Korean-language evaluation. The 3B variant anchors v0.1 alongside

Kanana 2.1B; the 7B variant tests whether scaling the multilingual baseline catches up to Korean-native specialists at similar parameter count.

- **Korean-tuned Llama 3 derivatives (Llama-3 Community License).** Blossom-8B and Open-Ko-8B are independent Korean fine-tunes of Meta Llama 3 8B, released by MLP-KTLim and [beomi](#) respectively. They test whether a Korean fine-tune of a stronger non-Korean base outperforms a Korean-native model at the same parameter count; the Llama-3 Community License permits research use and redistribution under attribution and naming rules that we preserve in the repository's [NOTICE](#) file.

The retriever is shared across all SUTs: [intfloat/multilingual-e5-small](#) (MIT, 118 M parameters), with mean-pooled L2-normalized embeddings and a top-5 cosine policy as described in §2.3.

2.5 MIRROR policies

We implement two policies in [src/mirror/prompts.py](#). The advisory policy serves as a generic-assistant baseline, permitting imperatives and targeting three to eight sentences of concise bullet advice. The reflective policy imposes three hard constraints: each response must cite at least one retrieved memory verbatim, avoid imperatives entirely, and end with an open question, within a three- to six-sentence target. Advisory is the baseline against which we measure the Part 1 MIRROR style; a third listening policy is implemented but unused in v0.1.

2.6 Conditions

We compare three memory conditions. The *no_node* baseline presents only the today-question. The *retrieval* condition augments the question with the top-5 neighbors from the embedding index. The *oracle* condition replaces retrieval with every document whose id appears in the item's [evidence_*_ids](#) lists, capped at $k \times 3$ (= 15) for prompt length. Algorithm 2 gives the test-time pipeline shared by all three.

Algorithm 2 – Retrieval-augmented inference

```

-----
input  : item x, SUT m, policy  $\pi \in \{\text{advisory}, \text{reflective}\}$ ,
        condition  $c \in \{\text{no\_node}, \text{retrieval}, \text{oracle}\}$ ,
        index I, embedder E, top-k = 5
output : response string r

if c = no_node:
    ctx  $\leftarrow \emptyset$ 
elif c = retrieval:
    q  $\leftarrow E.\text{encode}(x.\text{question})$ 
    ctx  $\leftarrow \text{top\_k}(I, q, k = 5)$            // cosine similarity
elif c = oracle:
    ctx  $\leftarrow \{ u \in I.\text{docs} \mid u.\text{id} \in x.\text{evidence\_ids} \}$ 
    ctx  $\leftarrow \text{ctx}[0 : 3 \cdot k]$            // cap at 15

prompt  $\leftarrow \text{format}(\pi, \text{ctx}, x.\text{question})$  // policy prompt template
r  $\leftarrow m.\text{generate}(\text{prompt}, \text{max\_tokens} = 200, \text{temperature} = 0.3)$ 
return r

```

2.7 Judging

Reflective-quality pairs (advisory vs reflective response) were scored by two cost-tier judges, `openai/gpt-4o-mini` and `anthropic/claude-haiku-4.5`, on four rubrics (specificity, non_directive, emotional_attunement, open_question) with A/B swap for position-bias mitigation. 6 REFL items × 2 SUTs × 2 conditions × 2 judges × 2 swaps = **96 pairwise decisions per rubric** aggregated in `experiments/.../judge_aggregate.json`. Algorithm 3 details the pairwise protocol.

```

Algorithm 3 – Pairwise reflective-quality judging
-----
input  : REFL items X_REFL, SUT pool S, judges J,
         rubrics R = {specificity, non_directive,
                     emotional_attunement, open_question}
output : win counts W[s, c, rubric] → {adv, refl, tie}

for each x ∈ X_REFL, s ∈ S, c ∈ {retrieval, oracle}:
    r_adv ← results[x, s, c, advisory].response
    r_refl ← results[x, s, c, reflective].response
    for each j ∈ J:
        for swap ∈ {False, True}:
            (A, B) ← (r_refl, r_adv) if swap else (r_adv, r_refl)
            v ← j.score(x.question, A, B, R) // per-rubric
            if swap:
                v ← {A↔B, tie↔tie} // invert
            for each p ∈ R:
                W[s, c, p][v[p]] += 1
return W

```

Each rubric therefore accumulates $|X_REFL| \cdot |S| \cdot 2 \cdot |J| \cdot 2 = 6 \cdot 2 \cdot 2 \cdot 2 \cdot 2 = 96$ decisions.

2.8 Budget and reproducibility

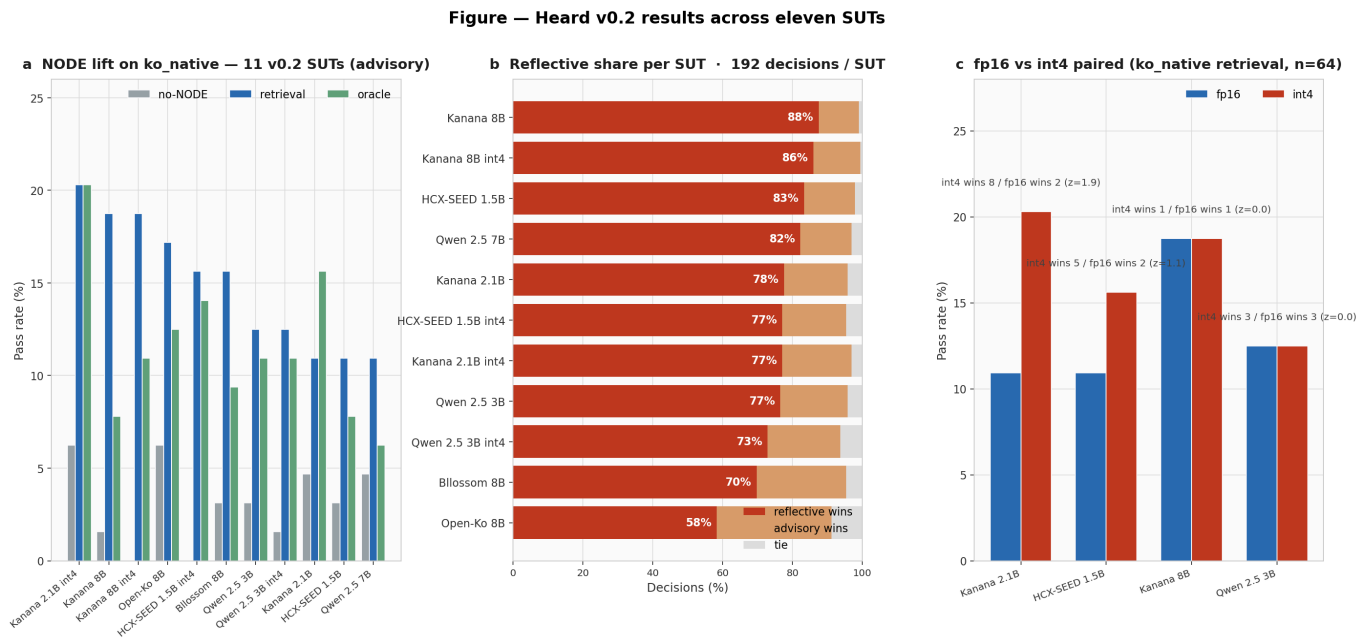
Phase	API cost (USD)
Day 1 utterance corpus	1.88
Day 2 scenarios + translation + 4-gate	2.04
Day 3 judge	0.06
Misc / smoke / retries	~0.02
Total	≈ 4.00 / 30.00 cap (13%)

Every API call logs (`model`, `temperature`, `seed`, `usage.cost`) to `experiments/_api_log/api_calls.jsonl`. Diskcache keyed on (`model`, `messages`, `temperature`, `seed`, `response_format`) makes a cold rerun free. Run artifacts (`config`, `results.jsonl`, `metrics`, per-SUT model revisions) are committed under `experiments/<run_id>/`.

3 Results

We report results from the v0.2 expanded sweep, which evaluates all eleven SUT configurations of Table A on the full v0.1 dataset under both advisory and reflective policies. Numbers are organized around the five

Tables 1–6 introduced below; Figure 2 visualizes the three primary findings.



{#fig:v02}

Figure 2. v0.2 results, three side-by-side panels. **Panel (a) — NODE lift on ko_native.** Eleven v0.2 SUTs, sorted left-to-right by retrieval pass rate, with grouped bars for the three memory conditions (no_node grey, retrieval blue, oracle sage). Every SUT records a positive retrieval–no_node delta; the strongest configuration (kanana_nano_int4) ties retrieval and oracle at 20.3 %. **Panel (b) — Reflective share per SUT.** Each horizontal bar aggregates 192 pairwise judge decisions per SUT (4 rubrics × 6 REFL items × 2 conditions × 2 judges × 2 A/B swaps); the red segment is reflective wins, the tan segment advisory, the light grey segment ties. Reflective share spans 58 % (open_ko_8b) to 88 % (kanana_8b). **Panel (c) — fp16 vs int4 paired.** For each of the four SUT families with both quantizations, the two bars show contains-pass on ko_native retrieval (n = 64 paired items); the annotation above each pair gives the item-level discordance — items where exactly one quantization passed — together with the McNemar z-score. Three of four pairs have $z \leq 1.13$; only Kanana 2.1B reaches a marginal $z = 1.90$.

3.1 NODE lift across the v0.2 lineup (Table 1)

The first headline result is that dense embedding retrieval lifts contains-pass on ko_native for every SUT in the lineup. Table 1 gives the per-SUT, per-condition pass rate under the advisory policy (n = 64 non-REFL items per cell, sorted by retrieval pass rate descending); Figure 2(a) is a graphical rendering of the same data.

Table 1. Advisory pass rate on ko_native, contains-token metric, n = 64. Sorted by retrieval.

SUT	no_node	retrieval	oracle	Δ retrieval–no_node
kanana_nano_int4	6.2%	20.3%	20.3%	+14.1 pp
kanana_8b	1.6%	18.8%	7.8%	+17.2 pp
kanana_8b_int4	0.0%	18.8%	10.9%	+18.8 pp
open_ko_8b	6.2%	17.2%	12.5%	+11.0 pp

SUT	no_node	retrieval	oracle	Δ retrieval-no_node
bllossom_8b	3.1%	15.6%	9.4%	+12.5 pp
hclova_seed_15b_int4	0.0%	15.6%	14.1%	+15.6 pp
qwen25_3b	3.1%	12.5%	10.9%	+9.4 pp
qwen25_3b_int4	1.6%	12.5%	10.9%	+10.9 pp
hclova_seed_15b	3.1%	10.9%	7.8%	+7.8 pp
kanana_nano	4.7%	10.9%	15.6%	+6.2 pp
qwen25_7b	4.7%	10.9%	6.2%	+6.2 pp

The retrieval lift is universal across the lineup. The mean **retrieval-no_node** delta is +11.8 pp and the minimum is +6.2 pp (**kanana_nano** and **qwen25_7b**); no configuration regresses with retrieval relative to the memoryless baseline. The Korean-native families (Kanana 1.5 and HyperCLOVA-X SEED) and the Korean-tuned Llama 3 derivatives populate the top half of the table; the multilingual Qwen 2.5 baselines remain mid-pack and never lead. The strongest single configuration is **kanana_nano_int4** at 20.3 % retrieval, followed by **kanana_8b** and **kanana_8b_int4** at 18.8 %. The on-device 2.1 B target named in Part 1 of the proposal — **kanana_nano** — sits at 10.9 % retrieval (4.7 % no_node, 15.6 % oracle), preserving the v0.1 headline trajectory of 4.7 % → 10.9 % → 15.6 % as a within-table reference point.

A second pattern in Table 1 is that retrieval and oracle no longer move in lockstep across the lineup. Five of eleven configurations record **oracle** strictly below **retrieval**, including the strongest large model **kanana_8b** (7.8 % oracle vs 18.8 % retrieval). The regression is concentrated in the 7–8 B SUTs and is absent in the smallest (**kanana_nano**, **hclova_seed_15b**); we read this as evidence that long oracle prompts (up to $k \times 3 = 15$ evidence documents) push larger SUTs into a long-context summarization mode that recapitulates the evidence rather than answering the question. We previously discussed this effect, on Qwen 2.5 3B alone, in §4.2 of v0.1; the v0.2 expansion shows it generalizes well beyond Qwen and is worth treating as a first-class artifact rather than a model-specific anomaly. A natural mitigation is an "answer-first" prompt guardrail; we leave it to v0.3 because its evaluation requires re-running the oracle column rather than re-aggregating existing data.

3.2 Quantization axis (Table 2; Figure 2c)

Four SUT families admit a 4-bit-NF4 quantization variant under the 30 GB memory budget; Table 2 contrasts their fp16 and int4 pass rates and latencies on the **ko_native** retrieval cell, and Figure 2(c) renders the corresponding paired item-level analysis. The aggregate reading from the marginal table is that the two smallest SUTs gain materially under int4 (Kanana 2.1B +9.4 pp, HyperCLOVA-X SEED 1.5B +4.7 pp) while the two larger SUTs are unchanged. We argue below that this aggregate reading overstates the int4 effect.

Table 2. *int4 vs fp16 on **ko_native** retrieval (advisory). Δ pass = int4 – fp16 in percentage points; latencies are mean seconds per generation across $n = 64$.*

SUT family	fp16 pass	int4 pass	Δ pass	fp16 latency	int4 latency
kanana_nano (2.1 B)	10.9 %	20.3 %	+9.4 pp	2.54 s	5.22 s
hclova_seed_15b (1.5 B)	10.9 %	15.6 %	+4.7 pp	1.83 s	3.45 s

SUT family	fp16 pass	int4 pass	Δ pass	fp16 latency	int4 latency
kanana_8b (8.0 B)	18.8 %	18.8 %	0.0 pp	3.73 s	3.60 s
qwen25_3b (3.0 B)	12.5 %	12.5 %	0.0 pp	2.66 s	4.81 s

A paired item-level analysis modulates the marginal +9.4 pp and +4.7 pp gains substantially. For each of the four SUT families, we count the items where fp16 and int4 disagree — items where exactly one of the two quantizations passed contains-token — and apply a McNemar two-sided z test. The discordance counts and z-scores reported in Figure 2(c) are: Kanana 2.1B int4 8 / fp16 2 ($z = 1.90$, $p \approx 0.06$), HyperCLOVA-X SEED 1.5B int4 5 / fp16 2 ($z = 1.13$), Kanana 8B int4 1 / fp16 1 ($z = 0$), Qwen 2.5 3B int4 3 / fp16 3 ($z = 0$). Only Kanana 2.1B reaches marginal significance, and even that result rests on a 6-item swing within a 64-item denominator with base pass rate near 15 %, where the standard error of a single configuration's pass rate is approximately ± 5 pp. The +4.7 pp HyperCLOVA-X SEED gain is not significant.

A second confound is decoding behavior. Mean response length on `ko_native` retrieval is 261 characters for `hclova_seed_15b` fp16 and 401 characters for `hclova_seed_15b_int4`; the int4 variant produces responses $\approx 1.5\times$ longer on average, while Kanana 8B and Qwen 3B produce responses of statistically indistinguishable length under the two quantizations (243 vs 226 chars and 244 vs 239 chars respectively). Because contains- token is a substring metric, longer responses are mechanically more likely to contain the gold tokens by coincidence; the HyperCLOVA-X SEED int4 gain is therefore at least partially a length artifact rather than evidence of better question answering. We do not have an analogous length explanation for the Kanana 2.1B gain (mean lengths 316 vs 322 chars), where the discordance is genuinely paired but borderline-significant.

The latency story is uncomplicated: int4 is *not* faster than fp16 on this hardware. Three of four SUT families show int4 latency 1.3–2.0 $\times$ longer than fp16, and only `kanana_8b` is neutral (3.60 s vs 3.73 s). The slowdown is consistent with the known dequantization overhead of bitsandbytes-NF4 on short generations — the per-token saving from the smaller weight matrix is offset by the per-generation fixed cost of unpacking 4-bit integers into half-precision compute — and is in line with the published benchmarks for this quantizer. For decision-moment assistants where the relevant deployment budget is GPU memory rather than wall-clock latency (an L40S can otherwise hold a fp16 8 B SUT and the retriever simultaneously, so the memory saving matters most below the L40S class), int4 remains an attractive option for the small Korean-native SUTs and a neutral option for the larger ones. We do *not* read the small-model contains-pass gains as evidence of better question answering under int4; they are best read as borderline-significant decoding artifacts in a 64-item benchmark.

3.3 Cross-track language axis (Table 4)

Table 4 reports advisory retrieval pass rate across the three heard-bench tracks for all eleven SUTs, generalizing the v0.1 single-SUT language-axis report. The denominators are the per-track non-REFL counts established in §2.2: 64 on `ko_native`, 100 on `en_subset` and `ko_translated`.

Table 4. Cross-track advisory retrieval pass rate, contains- token metric. Sorted by `ko_native`.

SUT	en_subset	ko_translated	ko_native
kanana_nano_int4	0.0 %	6.0 %	20.3 %
kanana_8b	2.0 %	5.0 %	18.8 %

SUT	en_subset	ko_translated	ko_native
kanana_8b_int4	0.0 %	2.0 %	18.8 %
open_ko_8b	2.0 %	2.0 %	17.2 %
bllossom_8b	2.0 %	4.0 %	15.6 %
hclova_seed_15b_int4	0.0 %	2.0 %	15.6 %
qwen25_3b	0.0 %	2.0 %	12.5 %
qwen25_3b_int4	0.0 %	1.0 %	12.5 %
hclova_seed_15b	0.0 %	1.0 %	10.9 %
kanana_nano	0.0 %	5.0 %	10.9 %
qwen25_7b	1.0 %	2.0 %	10.9 %

The v0.1 language-axis ordering $\text{en_subset} \leq \text{ko_translated} < \text{ko_native}$ is preserved by ten of eleven configurations (**open_ko_8b** ties `en_subset` and `ko_translated` at 2.0 %). Mean pass rates over the lineup are 0.6 % on `en_subset`, 2.9 % on `ko_translated`, and 14.8 % on `ko_native` — a 25× gap between the two endpoints. The multilingual Qwen 2.5 baselines do not close the `en_subset` gap despite English being their native training distribution, which we attribute to the LongMemEval haystack length (sessions span hundreds of utterances per item) and the absence of session-level retrieval in our pipeline; this is the v0.1 limitation we discuss in §4.4 and is unchanged in v0.2. A second observation is that `ko_translated` sits closer to `en_subset` than to `ko_native` across the board even though the question and gold answer are in Korean — confirming the ADR-0003 hypothesis that haystack language dominates for this kind of task and motivating a future **ko_translated_full** track that translates the haystack as well as the question.

3.4 Reflective policy: contains-token neutrality and pairwise dominance (Tables 3, 5, 6; Figure 2b)

We measure the reflective policy along two axes: contains-token accuracy (does the reflective response still surface the gold substring?) and the pairwise reflective-quality judge described in §2.7 (does it sound more like a mirror?). The contains-token view (Table 3) shows the policy is approximately accuracy- neutral; the pairwise view (Tables 5 and 6, Figure 2b) shows it dominates on the qualitative rubrics that operationalize the "mirror, do not advise" thesis.

Table 3. *Reflective – Advisory contains-pass on **ko_native** retrieval, percentage points.*

SUT	advisory	reflective	Δ (refl – adv)
bllossom_8b	15.6 %	28.1 %	+12.5 pp
kanana_nano	10.9 %	20.3 %	+9.4 pp
qwen25_7b	10.9 %	14.1 %	+3.1 pp
kanana_8b	18.8 %	21.9 %	+3.1 pp
kanana_8b_int4	18.8 %	18.8 %	0.0 pp

SUT	advisory	reflective	Δ (refl – adv)
qwen25_3b_int4	12.5 %	10.9 %	–1.6 pp
hclova_seed_15b	10.9 %	9.4 %	–1.6 pp
hclova_seed_15b_int4	15.6 %	12.5 %	–3.1 pp
qwen25_3b	12.5 %	9.4 %	–3.1 pp
kanana_nano_int4	20.3 %	17.2 %	–3.1 pp
open_ko_8b	17.2 %	10.9 %	–6.2 pp

Across eleven SUTs the mean contains-pass shift from advisory to reflective is +0.9 pp with median 0.0 pp, four positive deltas, two zero, and five negative; the spread (–6.2 to +12.5 pp) is within the noise band one would expect from a 64-item denominator. We read the policy as accuracy-neutral on the contains-token metric — the hard constraints (cite a memory verbatim, no imperatives, end with an open question) do not prevent the SUT from surfacing the gold substring, they merely route it through a different stylistic shell.

The pairwise judge measures the value of that shell directly. Each (item, SUT, condition) triple where both advisory and reflective responses exist is scored under two judges (*gpt-4o-mini*, *claude-haiku-4.5*) with A/B-swap symmetrization on each of four rubrics (specificity, non-directive, emotional attunement, open question). The v0.2 sweep yields 132 valid triples (eleven SUTs $\times$ six REFL items $\times$ two retrieval-augmented conditions; reflective $\times$ *no_node* is excluded by the runner because the two policies degenerate without retrieved context), and $132 \times 2 \times 2 \times 4 = 2,112$ verdicts in total. Table 5 sums verdicts across all SUTs and conditions per rubric; Table 6 sums across all conditions and rubrics per SUT ($n = 192$ decisions per SUT).

Table 5. Pairwise judge totals per rubric, summed across all eleven SUTs and both retrieval-augmented conditions. Each rubric column sums to $528 = 11 \text{ SUTs} \times 2 \text{ conditions} \times 6 \text{ REFL items} \times 2 \text{ judges} \times 2 \text{ A/B swaps}$.

Rubric	Advisory wins	Reflective wins	Tie	Reflective share
specificity	158	348	22	65.9 %
non_directive	178	350	0	66.3 %
emotional_attunement	32	465	31	88.1 %
open_question	33	466	29	88.3 %

Table 6. Reflective share per SUT, summed across both conditions and all four rubrics ($n = 192$ decisions per SUT). Sorted by reflective share.

SUT	Advisory wins	Reflective wins	Tie	Reflective share
kanana_8b	22	168	2	87.5 %
kanana_8b_int4	26	165	1	85.9 %
hclova_seed_15b	28	160	4	83.3 %
qwen25_7b	28	158	6	82.3 %

SUT	Advisory wins	Reflective wins	Tie	Reflective share
kanana_nano	35	149	8	77.6 %
kanana_nano_int4	38	148	6	77.1 %
hclova_seed_15b_int4	35	148	9	77.1 %
qwen25_3b	37	147	8	76.6 %
qwen25_3b_int4	40	140	12	72.9 %
bllossom_8b	49	134	9	69.8 %
open_ko_8b	63	112	17	58.3 %

The reflective policy wins every rubric by a clear margin; the two rubrics that operationalize the "mirror, do not advise" thesis — emotional attunement and open-question framing — land at 88.1 % and 88.3 % reflective share respectively, recovering the v0.1 numbers ($82/96 \approx 85\%$ and $88/96 \approx 92\%$) under a 5.5 $\times$ -larger sample. Specificity and non-directive remain tighter at 66 %; the advisory policy occasionally wins on specificity by virtue of being less constrained in surface form, and on non-directive by accident when the advisory generation phrases a recommendation as a question.

The per-SUT view is more discriminating than the per-rubric average. Reflective share spans 58.3 % ([open_ko_8b](#)) to 87.5 % ([kanana_8b](#)); every SUT has the reflective policy ahead, but the margin varies considerably. The two strongest SUTs on this metric are the Kanana 8B family (fp16 87.5 %, int4 85.9 %), and the two weakest are the Korean-tuned Llama 3 derivatives ([bllossom_8b](#) 69.8 %, [open_ko_8b](#) 58.3 %). We attribute the Llama-3-derivative weakness to a stronger advisory baseline rather than weaker reflective output: [bllossom_8b](#) and [open_ko_8b](#) accumulate the highest advisory-win counts in the lineup (49 and 63 respectively, against ≤ 40 elsewhere), indicating these models follow advisory imperative templates more willingly than the Kanana / Qwen / HyperCLOVA-X families. The reflective policy still wins, but it is competing against a more imperatively-styled advisory response and the margin narrows.

3.5 Latency and SUT recommendation

Table 7 summarizes mean per-generation latency on [ko_native](#) across all eleven SUTs and three conditions, on a single L40S 48 GB. The table is the v0.2 analogue of the v0.1 Pareto plot, but without the figure: with eleven SUTs in two parameter decades, a scatter plot is no longer a clean visual.

Table 7. Mean wall-clock latency per response (s) on [ko_native](#), advisory policy.

SUT	no_node	retrieval	oracle
hclova_seed_15b	1.78	1.83	1.78
kanana_nano	2.25	2.54	2.40
qwen25_3b	1.90	2.66	2.16
qwen25_7b	2.74	3.15	2.58
kanana_8b	3.25	3.73	3.27
kanana_8b_int4	3.17	3.60	3.29

SUT	no_node	retrieval	oracle
hclova_seed_15b_int4	3.37	3.45	3.37
open_ko_8b	3.41	3.87	3.63
qwen25_3b_int4	4.07	4.81	4.01
bllossom_8b	4.90	4.94	4.91
kanana_nano_int4	4.83	5.22	5.05

Latencies range from 1.78 s (`hclova_seed_15b` `no_node`) to 5.22 s (`kanana_nano_int4` retrieval). The lineup splits roughly into a sub-3 s tier (the Korean-native fp16 SUTs at 1.5–8 B) and a 3–5 s tier (most int4 variants and the Llama 3 derivatives). All SUTs remain within the rough on-device latency budget of 5–7 s for a decision-moment assistant operating at human conversational pacing.

Combining the four orthogonal v0.2 metrics — retrieval pass rate (§3.1), reflective share (§3.4), latency (this section), and quantization sensitivity (§3.2) — singles out `kanana_8b` as the v0.2 recommended SUT for the *Heard* product. It places second on retrieval pass rate (18.8 %, behind only `kanana_nano_int4`'s likely-noise 20.3 %), first on reflective share (87.5 %), in the mid-latency tier (3.73 s retrieval), and is one of the two SUT families where int4 quantization is exactly latency-neutral, so the int4 variant offers a clean memory trade for deployment on hardware below the 48 GB L40S class. The on-device 2.1 B target named in Part 1 — `kanana_nano` — remains the right choice for the lightest deployment envelope (1.83 s retrieval, 77.6 % reflective share), but at the 8 B class the in-family scale-up preserves all four properties.

4 Discussion

4.1 Absolute pass rates are low by construction

Adversarial filtering retained only items where three different no-memory baselines failed across three trials. The surviving items are, by construction, hard. Kanana's 10.9% retrieval pass rate on this subset is meaningful evidence of memory contribution; the same SUT on unfiltered questions would look much better on paper and teach less. The headline interpretation of §3.1 is the relative lift ($\times 3.3$ end-to-end), not the absolute rate.

4.2 The retrieval > oracle regression

Five of eleven v0.2 configurations record `oracle` strictly below `retrieval` on `ko_native` (Table 1), with the regression concentrated in the 7–8 B parameter tier (`kanana_8b`, `kanana_8b_int4`, `qwen25_7b`, `bllossom_8b`) and absent in the two smallest SUTs (`kanana_nano`, `hclova_seed_15b`). We read this as a long-context summarization bias: a 15-document oracle prompt approximately triples the prompt length of a 5-neighbor retrieval prompt, and the larger SUTs respond by recapitulating the evidence sessions rather than answering the today-question. A concrete fix is an "answer-first" guardrail in the system prompt — *"Answer the question. Do not re-narrate the past."* — which we expect to close most of the retrieval-vs-oracle gap on the affected SUTs. Validating that prediction is a v0.3 task because it requires re-running the oracle column rather than re-aggregating existing data; it reads as a prompt-design artifact rather than a capability ceiling.

4.3 What dense retrieval does not measure

The v0.1 5-ability benchmark measures substring recall, namely whether the SUT reproduces the correct string given retrieved evidence. Dense embedding retrieval handles this well: when the evidence session is in the top-5, substring recall usually succeeds.

Three axes fall outside what this benchmark scores cleanly, each connecting to a Part 1 NODE claim. First, absence queries of the form "the user has never mentioned a vegan menu" resist dense retrieval because cosine similarity always returns some neighbor, encouraging hallucinated confirmation rather than refusal. Second, *entity-aggregation* queries such as "every memory about Grandma Park" require exhaustive graph traversal for completeness, whereas cosine top-k is both approximate and bounded in coverage. Third, *temporal-latest* queries such as "the current permanent-wave product" need exact timestamp filtering, whereas semantic similarity conflates old and new mentions. A NODE-native benchmark of 30 to 50 items targeting these three axes is the natural v2 extension.

4.4 Track B scope limitation

ko_translated keeps the haystack in English (ADR 0003). The "language only" axis is thus partially confounded, because Kanana's ko_translated number reflects both translation and haystack-language effects. Re-running with Korean haystacks is a tractable extension; the cost analysis in ADR 0003 identifies chunked session translation as the right path.

4.5 Validation shortfall

The final ko_native set is 70 items, below the planned 100. Gate 4 (question clarity $\geq 3/5$) was the dominant rejection cause: LLM raters are systematically conservative on isolated questions shown without persona context. Giving the rater that context in a future run is a one-line prompt change likely to lift acceptance closer to what a human reviewer would accept.

4.6 What the int4 axis does and does not show

The marginal +9.4 pp and +4.7 pp gains on [kanana_nano_int4](#) and [hclova_seed_15b_int4](#) (Table 2) are easy to misread as evidence that 4-bit-NF4 quantization *helps* contains-token accuracy on small Korean-native SUTs. The paired item-level analysis in §3.2 and Figure 2(c) does not support that reading. Of the four fp16/int4 SUT pairs, three have McNemar z-scores at or below 1.13 (i.e., not significant at any conventional level), and the fourth (Kanana 2.1B) reaches $z = 1.90$, which is borderline significant in a 64-item denominator at base rate $\approx 15\%$. The HyperCLOVA-X SEED int4 gain is moreover confounded by a decoding-behavior shift: mean response length on [ko_native](#) retrieval grows from 261 characters (fp16) to 401 characters (int4) for that SUT alone, while the other three pairs produce responses of indistinguishable length under the two quantizations. Because contains-token is a substring metric, longer responses are mechanically more likely to contain the gold tokens by coincidence; we therefore treat the HyperCLOVA-X SEED int4 advantage as a length artifact rather than a quality gain.

We retain the int4 axis in the sweep because the latency and memory analyses still favor it for two specific deployment profiles. Three of four SUT families show int4 *latency* 1.3–2.0× longer than fp16 on the L40S (§3.2), so int4 is not a throughput optimization on this hardware; the relevant deployment profiles are memory-bound, where int4 quarters the weight memory of an 8 B SUT and lets it co-reside with the retriever on devices below the 48 GB L40S class, and [kanana_8b](#) specifically, where the int4 variant is exactly latency-neutral and incurs no contains-pass cost. We do not recommend int4 as a quality improvement, only as a memory option with a known small accuracy risk.

4.7 What the v0.2 expansion leaves unmeasured

The v0.2 sweep expands the SUT axis from two to eleven configurations and adds a quantization axis, but does not expand the dataset axis. The three v0.1 dataset limitations articulated in §4.3 (NODE-native abilities not measured by substring recall), §4.4 (English haystack on `ko_translated`), and §4.5 (70 items rather than the planned 100) are unchanged.

Three SUT-axis items also remained out of reach. The two Korean-tuned 10 B+ models we wanted to include — EEVE-Korean 10.8B and SOLAR 10.7B — were dropped from the sweep at runtime when their fp16 throughput proved too low to finish within the wall-clock budget (≈ 47 and ≈ 200 generations per hour respectively under the reflective policy); they remain v0.3 candidates once an int4 path with adequate generation speed is identified. EXAONE-3.5 2.4B remained excluded by a `transformers` import incompatibility (`RopeParameters`) that the v0.2 environment did not resolve. And a BM25 baseline that would isolate the contribution of dense retrieval against a sparse counterpart is not yet in the lineup.

5 Conclusion

Heard contributes, to our knowledge, the first Korean long-term memory benchmark targeting solo-business monologue, built because no existing benchmark meets the three criteria of Korean / monologue / solo-business. On this benchmark, the v0.2 sweep evaluates eleven SUT configurations across four model families and a 4-bit-NF4 quantization axis under both advisory and reflective response policies, generating 14,850 SUT generations and 2,112 pairwise judge verdicts.

Three results are robust across the lineup. First, dense retrieval lifts contains-pass on `ko_native` for every configuration (mean +11.8 pp, minimum +6.2 pp, no negative deltas), with the on-device 2.1 B target named in Part 1 (`kanana_nano`) following the canonical 4.7 % $\rightarrow$ 10.9 % $\rightarrow$ 15.6 % trajectory. Second, the reflective policy dominates the advisory baseline on every rubric across 528 pairwise judge decisions per rubric, with the two "mirror, do not advise" rubrics — emotional attunement and open-question framing — landing at 88.1 % and 88.3 % reflective share. Third, the language-axis ordering `en_subset` $\leq$ `ko_translated` $<$ `ko_native` is preserved by ten of eleven SUTs, with multilingual baselines unable to close the gap on Korean tracks. Combining the four v0.2 metrics (retrieval pass, reflective share, latency, int4 sensitivity) we recommend `kanana_8b` as the default *Heard* deployment SUT, with its int4 variant as a memory-bound alternative.

Two results require careful framing. The marginal int4 contains-pass gains on the two smallest SUTs (Kanana 2.1B +9.4 pp, HyperCLOVA-X SEED 1.5B +4.7 pp) reduce to one borderline-significant pair ($z = 1.90$) and one confounded by a 1.5 $\times$ response-length increase under paired item-level analysis; we explicitly do not read these as a quality improvement from quantization (§4.6). And five of eleven SUTs show retrieval > oracle regression on `ko_native`, concentrated in the 7–8 B parameter tier, which we attribute to long-context summarization bias and propose to fix with an "answer-first" guardrail in v0.3 (§4.2).

Several extensions follow from the limitations identified above, ordered by rough value-to-effort priority.

1. A NODE-native capability benchmark of 30 to 50 items targeting absence-strict, entity-aggregation, and temporal- latest queries, the three axes §4.3 argues dense retrieval cannot reach.
2. Full-haystack Korean translation of `ko_translated` via chunked session translation (ADR 0003), closing the confound in §4.4.
3. An "answer-first" prompt guardrail to test the §4.2 hypothesis that the retrieval > oracle regression on five of eleven SUTs is a long-context summarization artifact.

4. SUT expansion to EXAONE-3.5 (pending a `transformers` import fix) and to EEVE-Korean 10.8B / SOLAR 10.7B (pending an int4 path with adequate generation speed).
5. A BM25 baseline to isolate the dense-retrieval contribution against a sparse counterpart.

6 Artifacts and References

6.1 Artifacts

The author website, code repository, and dataset repository links appear in the header of this report. The remaining artifacts used for the results in §3 are located as follows.

Location	
v0.2 expanded sweep (merged, 11 SUTs)	<code>experiments/20260426_1242_v0.2_sweep_merged/</code>
v0.2 sweep results (raw)	<code>experiments/20260426_1242_v0.2_sweep_merged/results.jsonl</code>
v0.2 aggregate metrics	<code>experiments/20260426_1242_v0.2_sweep_merged/metrics.csv, metrics.json</code>
v0.2 judge verdicts	<code>experiments/20260426_1242_v0.2_sweep_merged/judge_verdicts.jsonl</code>
v0.2 judge aggregate	<code>experiments/20260426_1242_v0.2_sweep_merged/judge_aggregate.json</code>
v0.1 reference sweep (2 SUTs, archive)	<code>experiments/20260423_1610_day3_sweep/</code>
Per-call API log	<code>experiments/_api_log/api_calls.jsonl</code>
ADRs	<code>docs/decisions/0001-0004*.md</code>

6.2 References

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7 Acknowledgments and AI Use

Claude Code was used as a coding assistant for implementing the pipeline and scripts in this repository. Figure 1 ([fig_overview.jpg](#)) was drawn in Paperbanana. All design decisions, including research framing, dataset methodology, hypothesis formulation, result interpretation, and writing direction, were performed by the author.

Appendix A — System prompts used during dataset construction

Every dataset-building stage that calls an LLM does so under a short, fixed system prompt. The prompts are reproduced verbatim below for transparency; they live in the repository under [src/datagen/](#) and [src/eval/](#). Persona cards and per-call user messages are built around these system prompts at run time.

A.1 Utterance generator ([src/datagen/utterance_gen.py](#))

Used by [scripts/01_generate_data.py](#) to expand each (persona, day) block into 10–15 Korean nightly-self-talk utterances. Generators rotate across Haiku 4.5 / GPT-4o-mini / Gemini-2.5-flash.

당신은 한국 1인 자영업자의 **혼잣말 생성기**입니다.
페르소나 카드와 오늘의 이벤트를 보고, 그 사람이 실제로 내뱉을 법한 짧은 혼잣말을 만드세요.
규칙:

- (1) 가ㄱ 발화는 1-3문장, 구어체, 자연스러운 줄임말 허용.
- (2) 가ㅁ정·판단·의문이 섞여야 함. 뉴스 요약조 금지.
- (3) 가ㅌ은 날의 발화들은 주제가 이어지되 시점(아침·영업 중·마가ㅁ 후)이 달라야 함.
- (4) 이벤트의 topic_key 를 1개 이상 intended_topics 에 남겨주세요 – 나중에 retrieval gold label 로 씁니다.
- (5) 응답은 반드시 JSON 객체 {"utterances": [...]} 형식으로만.
- (6) 가ㄱ utterance 객체의 필드: time (HH:MM, 24h), text, intended_categories (customer/stock/pricing/mood/decision 중 해당되는 것), intended_topics (topic_key 리스트), references_historical_event (기억하는 과거 사건 힌트, 없으면 빈 리스트).

A.2 Scenario generator ([src/datagen/scenario_gen.py](#))

Used by [scripts/03_generate_scenarios.py](#) to compose 150 Track-C candidates from sampled evidence utterances. The prompt forbids fact invention and forces the gold answer to be derivable from the evidence already in the corpus.

당신은 한국어 long-term memory benchmark 의 **시나리오 생성기**입니다. 주어진 페르소나의 과거 발화(evidence)를 토대로, 오늘 시점의 질문 하나를 만듭니다.

사실을 **새로 지어내지 마세요** – 질문과 정답은 evidence 에서
도출되어야 합니다.

출력은 반드시 JSON 객체:

```
{
  "question_text": "오늘 페르소나가 스스로에게 던지는 질문",
  "question_timestamp_day": "55~60 사이의 정수",
  "question_timestamp_hhmm": "HH:MM",
  "gold_answer_text": "정답 문장 (가ㄴ결)",
  "gold_contains_tokens": ["정답에 반드시 포함되어야 할  
짧은 substring (1~3개)"],
  "gold_excludes_tokens": ["hallucination 탐지용 (선택)"],
  "evidence_summary": "왜 이 evidence 가 정답의 근거인지 1문장",
  "reasoning": "생성 근거 (짧게)"
}
```

A.3 Korean translator ([src/datagen/translate.py](#))

Used by [scripts/05_translate_track_b.py](#) for the Track B (KO question + gold answer) translation.
ADR 0003 records why the haystack stays in English.

You are a professional English→Korean translator. Translate the user's question and the gold answer into natural Korean. Preserve numbers, proper nouns, and relative time expressions faithfully. Use casual Korean (해요체). Inside JSON strings, do NOT insert literal double-quote characters – use Korean 「」 or single ' ' quotes if you need to quote something. Output ONLY a JSON object:

```
{"question_ko": "...", "gold_answer_ko": "..."}

```

A.4 Validation gate 1 — evidence-answer consistency

([src/datagen/validation.py](#))

Asks a third-party LLM whether the gold answer is derivable from the candidate's evidence alone. Demoted to advisory by ADR 0004, but the verdict is still recorded per item.

You are a strict grader. Given evidence utterances (past monologues of a Korean shop owner) and a question, decide whether the gold answer can be reasonably derived from the evidence alone. Reply ONLY with a JSON object:

```
{"consistent": true|false, "reason": "1 sentence"}.

```

A.5 Validation gate 4 — question clarity

([src/datagen/validation.py](#))

Rates an isolated Korean question on a 1–5 Likert scale. Threshold $\geq 3 / 5$ retained 70 of 122 items (§2.2).

You are a strict question-clarity grader. Rate how clear and unambiguous a Korean monologue question is on a 1–5 Likert scale (5 = perfectly clear, 1 = unanswerable).
Reply ONLY with JSON:
`{"clarity": <1-5 integer>, "reason": "1 sentence"}`.

A.6 Pairwise reflective-quality judge ([src/eval/judge.py](#))

Used by [scripts/08_run_judge.py](#) to score advisory-vs-reflective response pairs across four rubrics. The hard constraint is the ternary {"A","B","tie"} output; the post-hoc swap-and-invert step in Algorithm 3 mitigates position bias.

You are a strict grader of Korean reflective-dialogue quality.

You compare two candidate responses (A and B) to the same user's utterance (오늘의 고민). Rate which one is better on each of four rubrics, on a ternary scale: "A" / "B" / "tie".

Rubrics:

specificity	Does it cite a specific past event with detail?
non_directive	Does it AVOID imperatives / prescriptive advice?
emotional_attunement	Does it acknowledge the speaker's feeling?
open_question	Does it end with an open (not yes/no) question?

Output ONLY JSON:

```
{"specificity":"A","non_directive":"tie", ...}
```